MIL-E-82401(CG)
15 November 1965
SUPERSEDING
CGS-52P-5b

MILITARY SPECIFICATION

ENAMEL, EXTERIOR, VINYL-ALKYD, RED

1. SCOPE

1.1 Scope. - This specification covers exterior vinyl-alkyd enamel of one type and grade in the color of red. This enamel is for use by brush or spray over pretreatment wash primer conforming to Specification MIL-P-15328 and vinyl-red lead primer conforming to Specification MIL-P-23281(CG).

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

RR-S-366	- Sieves, Standard for Testing Purposes
TT-D-643	- Drier, Paint, Naphthenate, Liquid, Concentrated
TT-M-268	- Methyl Isobutyl Ketone (For Use in Organic Coatings)
TT-P-143	- Paint, Varnish, Lacquer, and Related Materials; Packaging, Packing, and Marking of
TT-P-445	- Pigment, Toluidine-Red-Toner, Dry
TT-X-916	- Xylene (For Use in Organic Coatings)
PPP-C-96	- Can, Metal, 28 Gage and Lighter

FSC 8010

(1)

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MIL-P-15328 - Primer, Pretreatment (Formula No. 117 for Metals)

MIL-T-19588(CG) - Toluene-Methyl Isobutyl Ketone Mixture

MIL-P-23281(CG) - Primer, Vinyl-Red Lead (For Brush or Spray)

STANDARDS

Federal

FED-STD-141 - Paint, Varnish, Lacquer and Related Materials; Methods of Inspection, Sampling and Testing

FED-STD-595 - Colors; (FOR) Ready-Mixed Paints

3. REQUIREMENTS

3.1 Ingredients. - The enamel shall consist of ingredients conforming to the applicable specifications in the proportions shown in Table I.

3.2 Quantitative requirements. - The vinyl-alkyd red enamel shall conform to the quantitative requirements given in Table II when tested in accordance with Table III (4.6.1).

TABLE I - FORMULA

INGREDIENTS	RED		
	Percent by weight	Pounds per 100 gallons	Gallons per 100 gallons
Toluidine red toner (TT-P-445)	9.26	70.5	5.95
Methyl isobutyl ketone (TT-M-268)	51.14	389.7	58.81
Lead naphthenate drier (Type I, TT-D-643)	.03	0.2	.02
Cobalt naphthenate drier (Type II, TT-D-643)	.01	0.1	.01
Manganese naphthenate drier Type III, TT-D-643)	.01	0.1	.01
Xylene (Grade B, TT-X-916)	5.28	40.2	5.63
Vinyl resin 1/	10.28	78.3	6.75
Alkyd resin (TT-R-266, Type I (Class C)	23.99	182.7	22.82
TOTAL	100.00	761.8	100.00

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1/ Vinyl resin. - The vinyl resin shall be a hydroxyl containing vinyl chloride-acetate copolymer. It shall contain 89.5 to 91.5 percent vinyl chloride, 5.3 to 7.0 percent vinyl alcohol and 2.0 to 5.5 percent vinyl acetate. The resin shall have a specific gravity of 1.38 to 1.40. The material shall be furnished as a powdered solid not less than 98 percent of which shall pass through a No. 20 sieve conforming to the requirements of Specification RR-S-366. An 18 percent solution of the resin in methyl isobutyl ketone shall be no darker than 1.0 gram of potassium dichromate in one liter of water.

NOTE: Composition by volume, as shown in the column "Gallons per 100 gallons," is included for information only.

TABLE II - QUANTITATIVE REQUIREMENTS

CHARACTERISTICS	RED	
	Minimum	Maximum
Pigment, percent by weight	8.8	9.9
Volatiles, percent by weight	61.5	65.5
Nonvolatile vehicle percent by weight	26.5	27.5
Weight per gallon, pounds	7.1	8.1
Water, percent by weight	---	0.5
Coarse particles, as residue retained on standard No. 325 (44 micron) sieve, percent by weight of paint	---	0.5
Viscosity, Krebs units	55	67
Fineness of grind, Hegman gage	5	---
Flash point (Pensky-Martens Closed Cup)	50°F	---

3.3 Qualitative requirements. - The enamel shall meet the following qualitative requirements:

3.3.1 Manufacture. - The component raw materials shall be mixed and ground as required to produce a product which is uniform, homogeneous, free from grit, entirely suitable for the purpose intended, and in full conformity to the requirements of this specification.

3.3.2 Odor. - The odor shall be normal for the volatiles permitted when tested as specified in 4.6.1.

3.3.3 Skinning. - Skinning shall be absent when tested as specified in 4.6.1.

3.3.4 Drying time. - The enamel shall set to touch in not more than 15 minutes and shall dry sufficiently hard for recoating within one hour, when tested as specified in 4.6.1.

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3.3.5 Spraying properties. - The enamel shall have satisfactory spraying properties and shall be free from running, sagging, fogging, dusting, floating, or mottling when applied by spray and tested in accordance with 4.6.2.

3.3.6 Adhesion. - The enamel shall show good adhesion when tested as specified in 4.6.3.

3.3.7 Color.

3.3.7.1 Red. - The color of the red enamel shall match color chip No. 21105 of Standard FED-STD-595 when tested as specified in 4.6.4.

3.3.8 Compatibility with thinner. - There shall be no evidence of incompatibility of the enamel as received when diluted with vinyl paint thinner as specified in 4.6.5.

3.3.9 Gloss. - The paint shall have the highest possible gloss for the formulation specified.

3.3.10 Storage stability. - The product shall be of stable color and shall be capable of being readily broken up with a paddle into a smooth uniform consistency and shall not liver, thicken, curdle, gel, or show hard caking for at least one year after date of manufacture.

3.3.11 Additional tests (chemical and/or physical) may be run if considered necessary to determine that the ingredients and proportions conform to Table II.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own commercial laboratory acceptable to the Government. Inspection records of the examination and tests shall be kept complete and made available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. - For purposes of sampling, inspection, and testing, a lot shall consist of all material offered for delivery at one time, manufactured as one batch.

4.3 Inspection. - The procedure for inspection shall be in accordance with 1.1, 1.2, 1.3, or 1.4 of Method 1011 of Standard FED-STD-141, as stipulated in the contract or order (see 6.2).

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4.4 Containers. - Sample filled containers (quart and gallon) selected on the basis of four out of each 100 in the lot shall be examined for defects of construction of the container and the closure, for unsatisfactory markings, and all other preparation for delivery requirements. Each sample filled container shall also be weighed to determine the amount of contents. Any containers having one or more defects, or under required fill, shall be rejected and if the number of defective containers exceeds two per hundred in the lot, the whole lot shall be rejected.

4.5 Sampling for tests. - The procedure for sampling for tests shall be in accordance with Method 1021 of Federal Standard 141.

4.6 Tests.

4.6.1 Standard tests. - Standard tests conducted according to the applicable test method indicated in Table III shall be performed on the samples selected in accordance with 4.5 to determine conformance with the requirements of this specification. Numbers in the "Requirements Paragraph" column refer to the applicable paragraph of this specification. Numbers in the "Test Method" column refer to the applicable test method of Standard FED-STD-141.

TABLE III

Tests	Method	Requirement Paragraph
Pigment content	4022	3.2 (Table II)
Volatile content	4041	3.2 (Table II)
Nonvolatile vehicle content	4053	3.2 (Table II)
Weight per gallon	4184	3.2 (Table II)
Water content	4081	3.2 (Table II)
Coarse Particles	4092	3.2 (Table II)
Viscosity	4281	3.2 (Table II)
Fineness of grind	4411	3.2 (Table II)
Flash point (Pensky-Martens Closed Cup)	4293	3.2 (Table II)
Odor	4401	3.3.2
Skinning	3022	3.3.3
Drying time	4061	3.3.4

4.6.2 Spraying properties. - Clean a steel panel 2 by 2 feet or larger, by mechanical wire brush, apply one coat of pretreatment wash primer conforming to MIL-P-15328, and allow to dry 15 minutes at 20 degrees Centigrade. Then apply one coat of vinyl-red lead primer conforming to Specification MIL-P-23281 and allow to air dry 45 minutes. Support the panel at a 60 degree angle from the vertical, and apply by spraying the enamel to be tested. Observe for satisfactory spraying properties in all respects with special reference to freedom from running, sagging, or fogging. Leave the panel in position for four hours, then observe for freedom from dusting, floating, or mottling.

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4.6.3 Adhesion. - Apply a second coat of enamel to be tested over the panel specified in 4.6.2 and allow to dry 24 hours at 20 degrees Centigrade. The film shall then be subjected to a knife test to determine whether the enamel exhibits good adhesion to the vinyl-red lead primer.

4.6.3.1 As an alternative to the knife test, make a cut at least one inch long through the combined enamel film to bare metal with a sharp razor blade and then press a piece of cellulose (Scotch) tape (3/4 inch width) firmly against the enamel film over the cut section so that the length of the tape is perpendicular to the cut and so that the cut extends beyond the edges of the tape. The tape should be in contact with the enamel film for a distance of at least one inch on each side of the cut, and there should be sufficient excess tape on one side to be held firmly between the thumb and forefinger. Make the test by pulling the tape free in a slow and even manner in a direction at right angles to the panel. Not more than 1/8 inch of the film shall be picked up by the tape.

4.6.4 Color. - A panel for test shall be prepared by drawing down a coat of enamel to be tested with a doctor blade of 0.006-inch gap clearance to yield a wet-film thickness of approximately 0.003-inch on a plane piece of opaque white carrara glass. Forty-eight hours drying time under prevailing laboratory conditions shall be allowed. Compare the panel with the standard color chip for the color being tested as described in Method 4250 of Standard FED-STD-141.

4.6.5 Compatibility. - Compatibility with thinner shall be determined in accordance with Method 4203 of Standard FED-STD-141. Fifty milliliters of thinner conforming to Specification MIL-T-19588(CG) and fifty milliliters of enamel shall be used. Observations shall be made immediately after mixing and again 30 minutes after mixing.

4.6.6 Storage stability. - To determine conformance with 3.3.10 the enamel shall be checked by the Government inspector in accordance with Method 4142 of Standard FED-STD-141. Storage time shall be twelve months. Acceptance of a given lot of enamel need not await the completion of this test.

5. PREPARATION FOR DELIVERY

5.1 Packaging. - The enamel shall be packaged in 1-quart and 1-gallon containers, as stipulated (see 6.2). Individual 1-quart or 1-gallon containers shall conform to Type V, Class 2, of Specification PPP-C-96, except no exterior protective coating will be required.

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5.2 Marking.

5.2.1 Containers shall be marked in accordance with Specification TT-P-143 for military agencies, with the following added data: contract numbers and Federal stock number(s). Also, marking pertaining to application instructions shall be as follows:

This paint drys for recoating in 45 minutes at temperatures above 60 degrees Fahrenheit. Lower temperatures may require slightly longer drying time. For application instructions, see Coast Guard Paint and Color Manual (CG-263) Article 4-2-3.

PRECAUTIONS

Normal ventilation is always necessary. When air temperature exceeds 40 degrees Fahrenheit, take extra precautions to insure sufficient ventilation for low vapor concentration.

Enforce precautions to keep open flames, sparks cutting and welding operations away from painting and mixing areas.

5.2.2 Warning Label. - Each container shall have a warning label of appropriate size similar to Class 2 of MIL-L-19868 or shall be lithographed or stenciled with a reasonable likeness thereof. Under "contains" the solvent composition shall be inserted; methyl isobutyl ketone, xylene. For unit containers that also serve as shipping containers any conflict with ICC Regulations shall be resolved by reasonable modification of size of label or use of warning statement with label design.

5.2.3 Shipping containers. - Unless otherwise stipulated, shipping containers shall be marked with the name of the material, specification number, color, quantity and size(s) of the containers therein, the name of the contractor, the number of the contract or order, date of manufacture (month, year), and the gross weight.

6. NOTES

6.1 Intended use. - This specification covers an enamel which is intended for use as an exterior topcoat in the vinyl paint system, and is the standard red topcoat for buoys. It is designed for application on metal surfaces which have been treated with pretreatment, wash primer (MIL-P-15328), and coated with vinyl-red lead primer (MIL-P-23281). It will also adhere firmly to age-hardened conventional paint without lifting off the old paint film. It may be applied by spray or brush.

6.2 Ordering information. - Invitation for bids, contracts, and purchase orders should specify the following:

(a) Title, number and date of this specification
(b) Stock number(s)
(c) Color
(d) Inspection procedure (see 4.5)
(e) Size of containers and level of packing (see 5.1)

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6.3 Bid samples. - Unless otherwise specified, samples with bids are not required.

6.4 Vinyl-alkyd enamel should be purchased by volume, the unit being a U. S. gallon at 15.5 degrees Centigrade (60 degrees Fahrenheit).

6.5 Supersession data. - This specification includes the requirements of Coast Guard Specification CGS-52P-5b, dated 14 January 1953. The white, black and orange vinyl-alkyd paint previously procured under CGS-52P-5b is not procured under Specifications MIL-E-16738, MIL-E-15932 and TT-E-489 respectively.

NOTICE. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise, as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Preparing Activity:
U. S. Coast Guard

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SPECIFICATION ANALYSIS SHEET

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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